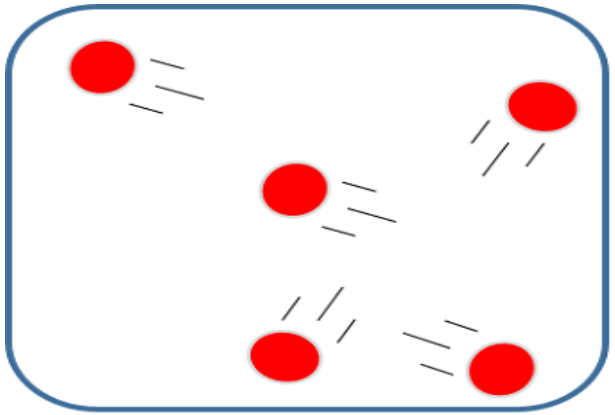
Fizika

9 sinf

MAVZU: Ideal gaz.

Ideal gaz

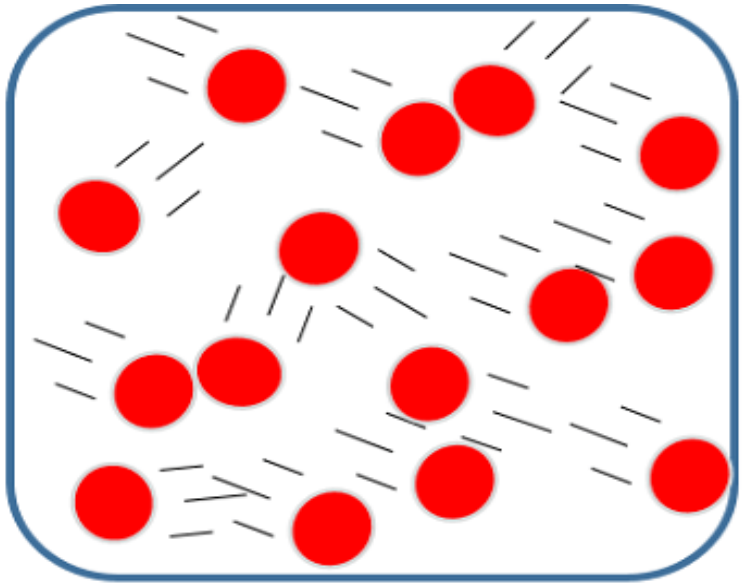
Molekulalari moddiy nuqta deb qaraladigan hamda ular orasidagi o‘zaro ta’sir kuchi e’tiborga olinmaydigan darajada kichik bo‘lgan gaz ***ideal gaz*** deb ataladi.



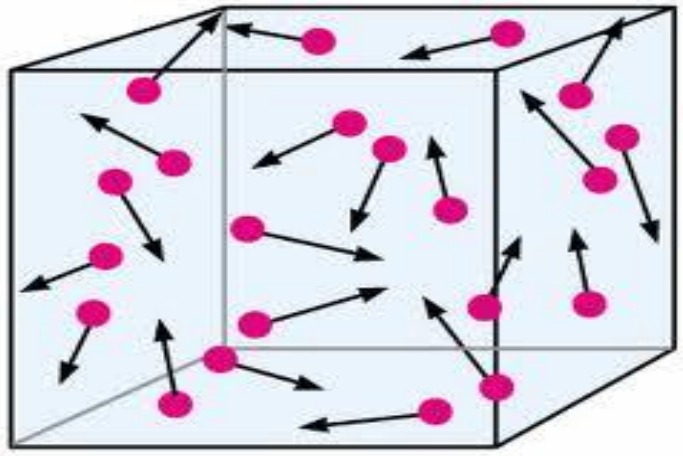
Real gaz

Tabiatda mutlaq ideal gazlar uchramaydi. Mavjud bo‘lgan gazlar ***real gazlar***dir.

Xossalari molekulalarning o‘zaro ta’sir kuchiga bog‘liq bo‘lgan gazlar ***real gazlar*** deb ataladi.



Ideal gazning bosimi



$$\vartheta_{o'rta}^2 = \frac{\vartheta_1^2+\vartheta_2^2+\cdots+\vartheta_N^2}{N}$$

Gazning bosimi gaz molekulalarining idish devoriga urilishi natijasida unga impuls ($m_0\vec{\vartheta}$) berilishi tufayli vujudga keladi.

$$p = \frac{1}{3} \cdot n \cdot m_0 \cdot \vartheta_{o'rt}^2$$

$n - gaz\ molekulalarining\ konsentratsiyasi$

$$p = \frac{1}{3} \cdot n \cdot m_0 \cdot \vartheta_{o'rt}^2 \Rightarrow p = \frac{2}{3} \cdot n \cdot \frac{m_0 \cdot \vartheta_{o'rt}^2}{2}$$

o‘rtacha kinetik energiya: $\overline{E}_k = \frac{m_0 \cdot \vartheta_{o'rt}^2}{2}$ dan

$$p = \frac{2}{3} \cdot n \cdot \overline{E}_k$$

$n \cdot m_0 = \frac{N}{V} m_0 = \frac{m}{V} = \rho$ dan $\left(n = \frac{N}{V}\, m = m_0 \cdot N\right)$

$$p = \frac{1}{3} \rho \cdot \vartheta_{o'rt}^2$$

Gazlar molekulyar-kinetik nazariyasining asosiy tenglamalari.

4-mashq 1-masala

Idish vodorod gazi bilan to‘ldirilgan. Idishdagi gaz molekulalarining konsentratsiyasi $4{,}5 \cdot 10^{24}\, m^{-3}$ ga teng. Idishdagi gazning bosimini hisoblang. Gaz molekulalarining o‘rtacha kvadratik tezligi 400 m/s ga teng deb oling.

Berilgan:

$$n = 4{,}5 \cdot 10^{24} m^{-3}$$

$$\overline{\vartheta} = 400\ m/s$$

$$M_{(H_2)} = 2\frac{g}{mol} = 2 \cdot 10^{-3}\frac{kg}{mol}$$

$$N_A = 6 \cdot 10^{23}\ mol^{-1}$$

Formula:

$$p = \frac{1}{3} \cdot n \cdot m_0 \cdot \overline{\vartheta^2}$$

$$m_0 = \frac{M}{N_A}$$

$$p = \frac{1}{3} \cdot n \cdot \frac{M}{N_A} \cdot \overline{\vartheta^2}$$

Topish kerak: $p-?$

Yechish:

$$p = \frac{1}{3} \cdot 4{,}5 \cdot 10^{24} m^{-3} \cdot \frac{2 \cdot 10^{-3}\frac{kg}{mol}}{6 \cdot 10^{23} mol^{-1}} \cdot \left(4 \cdot 10^2 \frac{m}{s}\right)^2 = 800\ Pa$$

Javob: P=800 Pa

4-mashq 3-masala

Idishdagi zichligi 1,5 kg/m^3 va bosimi 7,2 kPa bo‘lgan gaz molekulalarining o‘rtacha kvadratik tezligi nimaga teng?

Berilgan:

$\rho = 1{,}5\ kg/m^3$

$p = 7{,}2\ kPa = 7200\ Pa$

Topish kerak: $\overline{\vartheta} - ?$

Formula:

$$p = \frac{1}{3} \cdot n \cdot m_0 \cdot \overline{\vartheta^2}$$

$$n \cdot m_0 = \rho$$

$$p = \frac{1}{3} \cdot \rho \cdot \overline{\vartheta^2} \Rightarrow \overline{\vartheta^2} = \frac{3p}{\rho}$$

$$\overline{\vartheta} = \sqrt{\frac{3p}{\rho}}$$

Yechish: $$\overline{\vartheta} = \sqrt{\frac{3 \cdot 7200\ Pa}{1{,}5\ kg/m^3}} = 120\ m/s$$

Javob: $\overline{\vartheta} = 120\ m/s$

4-mashq 7-masala

30 kPa bosimda bir atomli gaz molekulasining o‘rtacha kinetik energiyasini toping. Berilgan bosimda bu gaz molekulalarining konsentratsiyasi $4 \cdot 10^{25}\, m^{-3}$ ga teng.

Berilgan:

$p = 30\ kPa = 3 \cdot 10^4\ Pa$

$n = 4 \cdot 10^{25}\ m^{-3}$

Formula:

$$p = \frac{2}{3} \cdot n \cdot \overline{E_k}$$

$$\overline{E_k} = \frac{3 \cdot p}{2 \cdot n}$$

Topish kerak: $\overline{E_k} - ?$

Yechish: $\overline{E_k} = \frac{3 \cdot 3 \cdot 10^4\ Pa}{2 \cdot 4 \cdot 10^{25}\ m^{-3}} = 1{,}125 \cdot 10^{-21}\ \text{J}$

Javob: $\overline{E_k} = 1{,}125 \cdot 10^{-21}\ \text{J}$

Mustaqil bajarish uchun topshiriqlar

1. Mavzuga doir savollarga javob yozish (20-sahifa).
2. 4-mashq 2-,4-,6-masalalarni yechish (20-sahifa).